fewer unnecessary values need to be read (that then are immediately discarded, as they don't contribute to an answer).

Further, column-oriented design allows you to apply storage optimization techniques, such as differential storage, like the XOR compression Facebook developed in the context of the Gorilla TSDB (see Tuomas Pelkonen et. al., "Gorilla: A Fast, Scalable, In-Memory Time Series Database." Proceedings of the VLDB Endowment, Vol. 8, No. 12, 2015; see also http://www.vldb.org/pvldb/vol8/p1816-teller.pdf) and run-length encoding (RLE; see D. Abadi, S. Madden, M. Ferreira, "Integrating Compression and Execution in Column-Oriented Database Systems." Proceedings of the 33th International Conference on Management of Data [SIGMOD]; https://doi.org/10.1145/1142473.1142548), yielding a smaller footprint and less resource usage.

In addition to compression, columnar design also enables vectorized execution (multiple data items per CPU cycle) and effective partitioning in the context of a single machine, and sharding a replication for scaling out, allowing you to answer queries over hundreds of millions of rows in subsecond time frames.

Prominent examples of columnar data stores, columnar formats, or services that use this strategy include the following:

- *Apache Cassandra (https://cassandra.apache.org/)*—Written in Java—a C++ variant, ScyllaDB, is also available (https://www.scylladb.com/).
- *Apache Druid (https://druid.apache.org/)*—An open source real-time analytics database with observability use cases by Airbnb, Alibaba, eBay, Lyft, and many more. You can run Druid in Kubernetes (http://mng.bz/0KDp).
- *Apache Pinot (https://pinot.apache.org/)*—An open source real-time distributed OLAP data store used by LinkedIn (http://mng.bz/KeOP), Uber, Slack, Stripe, and many more companies.
- *ClickHouse (https://clickhouse.com/)*—An open source columnar database, originally developed in 2008 by Yandex. You can use ClickHouse for all signal types, both for a single machine and in various distributed settings (sharded and replicated data with Apache Zookeeper as coordinator). When running it on Kubernetes, you should consider using the Altinity Operator for ClickHouse (https://github.com/Altinity/clickhouse-operator). ClickHouse has more-than-decent adoption, including Uber (logs), Cloudflare, and SigNoz, where it is being used as a building block in o11y solutions.
- *FrostDB (https://github.com/polarsignals/frostdb)*—A columnar database allowing for semistructured schemas, using Apache Parquet for storage as well as Apache Arrow for queries, developed as open source by Polar Signals.
- *Snowflake (https://www.snowflake.com/)*—An integrated platform delivered as a service. It features storage, compute, and global services layers that are physically separated but logically integrated.

If you're interested in learning more about the implementation details of some of the aforementioned data stores, I recommend reading the excellent article, "Comparison